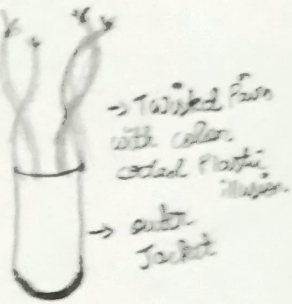
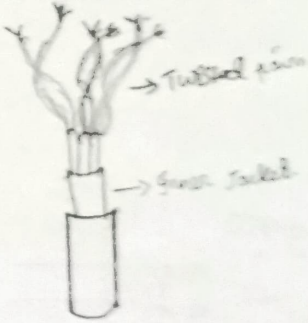
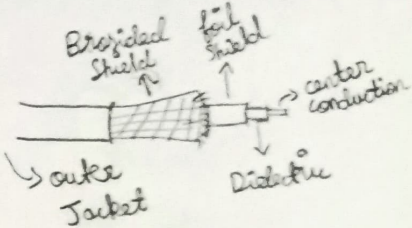
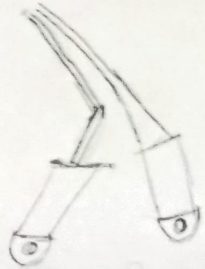
Exp: 2 Date:	Study of different types of Network Cables.

a. understand different types of network cables.

Different types of cables used in networking are:

1) unshield Twisted pair (UTP) cable
2) Shield Twisted pair (STP) cable
3) Coaxial cable
4) Fibre optic cable.

Category	Maximum Data Transmission	Advantage / Disadvantage	Application / use	Image
Category 3	10 bPS.	**Advantage** :- * Cheaper in cost. * Easy to install as they have a smaller overall diameter. **Disadvantage**:- * More prone to (EMI) electromagnetic interference and noise	10 Base-T Ethernet Fast Ethernet, Gigabit Ethernet. Fast Ethernet, gigabit ethernet.	
Category 5	upto 100 mbPS			
Category 5e	1 GbPS			
Category 6, 6a	10 GbPS.	**Advantage**:- * Shielded * Faster than UTP * Less susceptible to noise and interference. **Disadvantage**: * Expensive * Greater installation effort.	Gigabit Ethernet, 10G ethernet (55 m) widely used in data centres. Gigabit Ethernet, 10G Ethernet (100 m).	
Category 7	10 GbPS.			

Coaxial Cable	RG-6 RG-59 RG-11	10-100 MbPs	Advantage: * High bandwidth * Faster than UTP * Less susceptible to noise and interference * versatile Disadvantages:- * Limited distance * cost * Size is bulky.	Speed of signal is 500m television network High speed internet connections.	
fibre optics cable	Single mode multi mode	100 GbPs	Advantages: * High speed * High bandwidth * High security * Long distance Disadvantage: * Expensive * Requires skilled installers	Maximum distance of fibre optics cable is around 100 meters.	

Result:

The types of network cables and making cross wired cable & straight through cable using clamping/crimping tool is studied.